

● HARD

● GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

01

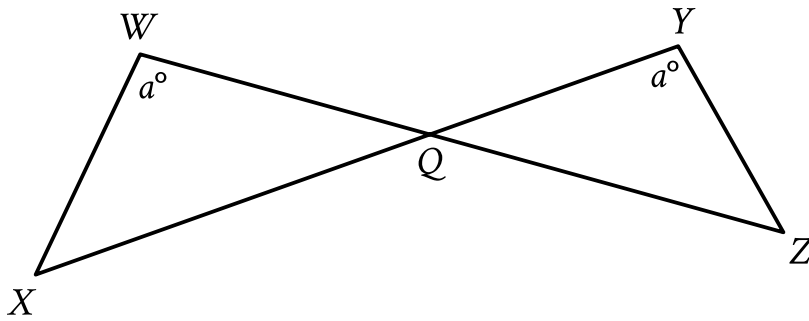
10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $4, 5$, $4, 7$, and $6, 5$, respectively. Triangle LMN has vertices L , M , and N at $4, 5$, $4, 7 + k$, and $6 + k, 5$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

- A. $90 - t - k^\circ$
- B. $90 - t + k^\circ$
- C. $90 - t^\circ$
- D. $90 + k^\circ$

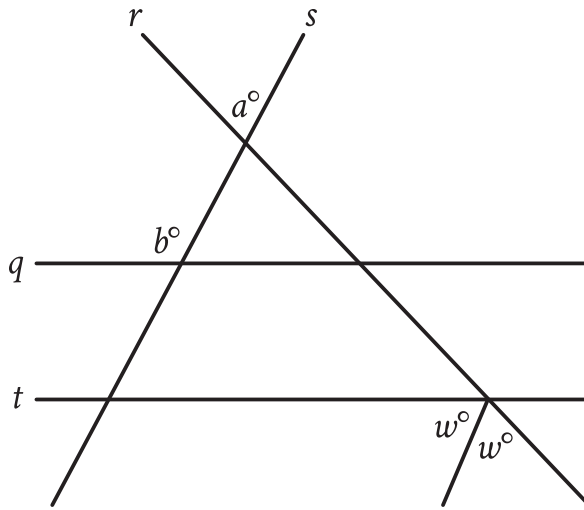


Note: Figure not drawn to scale.

- Triangle WXQ has a common vertex with triangle YZQ .
- The common vertex is Q .
- For triangle WXQ :
 - The measure of angle W is a° .
- For triangle YZQ :
 - The measure of angle Y is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A. 18
- B. 36
- C. 120
- D. 200



Note: Figure not drawn to scale.

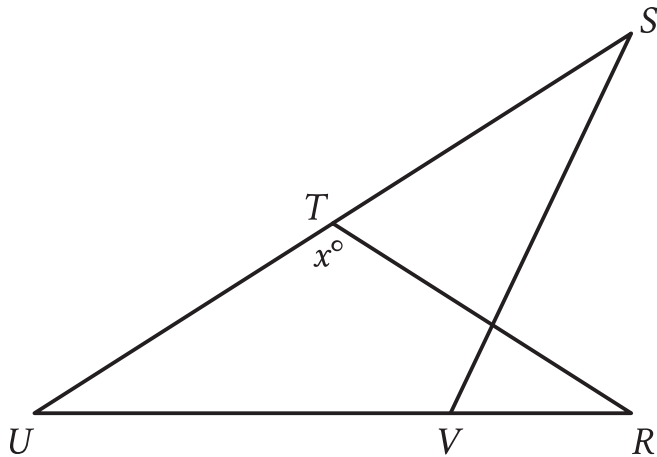
- Counterclockwise from top right, the lines are labeled s , r , q , and t .
- Line s intersects line r above line q .
- Line s and line r each intersect line q and line t .
- At the intersection of line s and line r , 1 angle is labeled counterclockwise from top as follows:
 - Top: a°
- At the intersection of line s and line q , 1 angle is labeled counterclockwise from top-left as follows:
 - Top left: b°
- At the intersection of line r and line t :
 - The line segment divides the bottom-left angle into two equal angles, each labeled w° .
- A note indicates the figure is not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

STUDENT-PRODUCED RESPONSE

	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Triangle $\triangle RST$ is labeled as follows:
 - The measure of angle $\angle RST$ is x° .
- The 2 triangles partially overlap:
 - Side RS of triangle $\triangle RST$ intersects side SV of triangle $\triangle TUV$.
 - Vertex V of triangle $\triangle TUV$ lies on side RU of triangle $\triangle RST$.
 - Vertex T of triangle $\triangle RST$ lies on side SV of triangle $\triangle TUV$.
- A note indicates the figure is not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

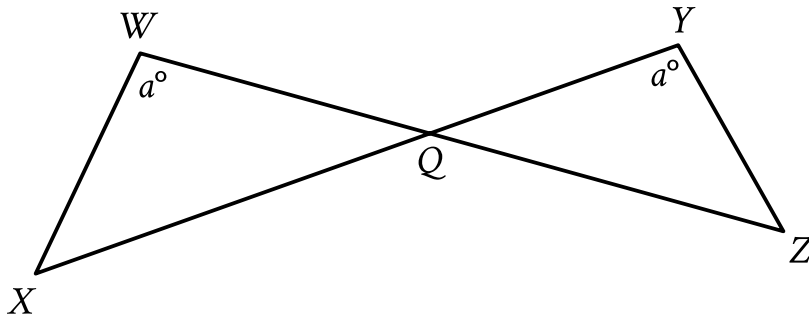
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GEOMETRY AND TRIGONOMETRY

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

- A. 5°
- B. 27°
- C. 45°
- D. 135°



Note: Figure not drawn to scale.

- The 2 triangles intersect at vertex upper Q.
- The measure of angle upper X upper W upper Q is a° .
- The measure of angle upper Z upper Y upper Q is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q. $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GEOMETRY AND TRIGONOMETRY

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

In triangle XYZ , angle Y is a right angle, point P lies on $\overline{XZ}$, and point Q lies on $\overline{YZ}$ such that $\overline{PQ}$ is parallel to $\overline{XY}$. If the measure of angle XZY is 63° , what is the measure, in degrees, of angle XPQ ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

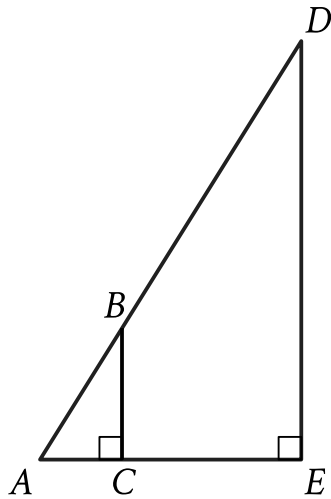
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper D upper E upper A is a right angle.
- Point upper B lies on side upper A upper D.
- Point upper C lies on side upper A upper E.
- Line segment upper B upper C is drawn from side upper A upper D to side upper A upper E to form right triangle upper A upper B upper C.
- Angle upper B upper C upper A is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)
(5)	(5)	(5)	(5)

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.